EE-312L Signals and Systems Lab
Laboratory Session 10
To understand and verify the Properties of Continuous-Time Fourier Transform using MATLAB

December 12, 2024

1 Objective

The objective of this lab session is to understand and verify the properties of Continuous-Time Fourier Transform using MATLAB.

2 Convolution in Time and Frequency [PV10]

A CT signal $x_1(t)$ and $x_2(t)$ and their respective Fourier transform signals, $X_1(j\omega)$, and $X_2(j\omega)$.

$$\mathcal{F}\{x_1(t) * x_2(t)\} = X_1(j\omega)X_2(j\omega) \tag{1}$$

$$\mathcal{F}\{x_1(t)x_2(t)\} = \frac{1}{2\pi}X_1(j\omega) * X_2(j\omega) \tag{2}$$

Example 01:

Let $x_1(t) = u(t) - u(t-2)$ and $x_2(t) = u(t) - u(t-4)$. Verify both sides of equation (1)

```
% Rigth Hand Side
clc;
clear all;
close all;
syms t w
x1 = heaviside(t)-heaviside(t-2);
x2 = heaviside(t)-heaviside(t-4);
```

```
X1 = fourier(x1,w);
X2 = fourier(x2,w);
right = ifourier(X1*X2,t);
%ezplot(right,[0 8]);
```

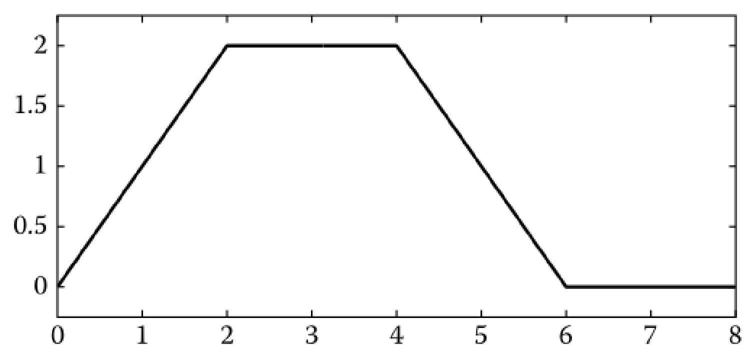

Figure 1: Computation and plot of the right side of (1)

```
% Left Hand Side
clc;
clear all;
close all;

t1 = 0:.01:2;
t2 = 2.01:.01:4;
x1 = [ones(size(t1)) zeros(size(t2))];
x2 = ones(size([t1 t2]));
y = conv(x1,x2)*.01;
figure(2)
plot(0:.01:8,y);
```

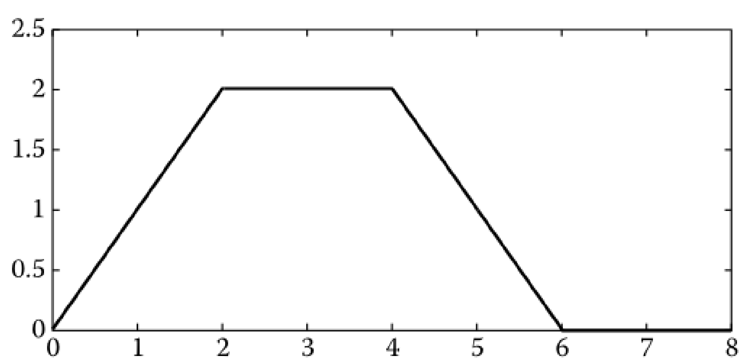

Figure 2: Computation and plot of the left side of (1)

3 Verification of Parseval's Theorem [PV10]

The Parseval's theorem is given in equation (3) as follows

$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(j\omega)|^2 d\omega \tag{3}$$

Example 02:

```
% Left Hand Side of Parseval's Theorem
clc;
clear all;
close all;

syms t w;
x = exp(-t^2);
Et = int( (abs(x))^ 2,t,-inf,inf);
eval(Et)
```

Result of left hand side : ans = 1.2533

```
% Right Hand Side of Parseval's Theorem
clc;
clear all;
close all;

syms t w;
x = exp(-t^2);
X = fourier(x,w);
Ew = (1/(2*pi))*int( (abs(X))^2,-inf,inf);
eval(Ew)
```

Result of right hand side : ans = 1.2533

4 Lab 10: Lab Tasks

Let $h(t) = e^{-t}u(t)$ and $x(t) = e^{-t}\cos(2\pi t)u(t)$. Verify both sides of equation (1).

References

[PV10] Alex Palamides and Anastasia Veloni. *Signals and systems laboratory with MATLAB*. CRC press, 2010.